

Insurance Claims & Portfolio Risk Dashboard

Data Analysis & Business Intelligence Track - Python + Tableau

Project starts	Saturday, August 22, 2026
Project work period	August 22 - September 1, 2026
Project upload days	Monday, August 31 & Tuesday, September 1, 2026
Final deadline	Tuesday, September 1, 2026
Feedback, corrections & video	Wednesday, September 2 - Friday, September 4, 2026

1. Business Scenario

You have joined the analytics team of a motor-insurance company. Management wants a clear view of the portfolio: where claims occur most frequently, where claim costs are highest, which policy segments deserve closer attention, and which patterns should be monitored in the next reporting cycle. Your job is to prepare the data in Python and build an executive Tableau dashboard that supports business discussion.

This is a descriptive analytics / BI project, not a pricing or underwriting model. Do not label a group as 'high risk' only because it has many claims; exposure and portfolio size must be considered.

2. Public Insurance Dataset

Use the public French Motor Third-Party Liability (freMTPL2) datasets from CASdatasets. The data describe 677,991 motor third-party liability policies observed mainly during 2011-2013. The frequency table contains policy/risk features and claim counts, while the severity table contains claim amounts linked by policy ID. The source is an unknown private insurer.

Official dataset documentation: <https://dutangc.github.io/CASdatasets/reference/freMTPL.html>

CASdatasets package index: <https://dutangc.github.io/CASdatasets/reference/index.html>

Important: freMTPL2 does not include written premium. Therefore, Loss Ratio must not be calculated from this dataset. Instead, use claim frequency, claim severity and total claim cost as the core insurance KPIs.

Key variables

IDpol - policy ID
 ClaimNb - number of claims during exposure
 Exposure - exposure period in years
 VehPower - vehicle power
 VehAge - vehicle age
 DrivAge - driver age
 BonusMalus - bonus/malus score
 VehBrand - vehicle-brand category
 VehGas - fuel type
 Area - density category from rural to urban
 Density - population density
 Region - policy region
 ClaimAmount - claim cost in the severity table

3. What You Should Know / Read

Python/Pandas: read data, clean types, groupby, agg, merge, sort_values, cut/qcut, handling duplicates and missing values.

EDA: distributions, skewed monetary values, outliers, segment comparison.

Insurance KPIs: policy count, exposure, claim count, claim frequency, average claim severity, total claim cost.

BI: KPI design, filters, drill-down, dashboard layout, business storytelling.

Tableau: data source connection, calculated fields, parameters/filters, bar/line/scatter charts, highlight tables and dashboard actions.

Required tools: Python, Jupyter/Colab or VS Code, pandas, NumPy, Matplotlib (optional for EDA), Tableau Public/Desktop, Git/GitHub.

4. Questions from the Insurance Manager

Q1 - What is the overall health of the insurance portfolio?

Build a portfolio summary with policy count, total exposure, total claims, claim frequency, total claim cost and average claim severity. Explain what each KPI means and why raw claim counts alone are misleading.

Q2 - Which regions generate the greatest claim burden?

Compare regions using exposure, claim frequency, total claim cost and average severity. Do not rank regions only by total claim count. Identify one region that deserves attention and justify the choice with at least two KPIs.

Q3 - Which driver and vehicle segments show different claim patterns?

Create sensible age/vehicle groups and compare claim frequency and severity across driver age, vehicle age, vehicle power, fuel type or bonus-malus categories. Explain the strongest patterns without claiming causation.

Q4 - Are frequent-claim segments also expensive-claim segments?

Compare claim frequency against average severity. Build a scatter plot or quadrant view that distinguishes high-frequency / high-severity segments from other combinations.

Q5 - Is claim cost concentrated in a small number of claims or policies?

Perform a Pareto-style analysis of ClaimAmount. Report what share of total claim cost is generated by the highest-cost claims/policies and explain why this matters for claims management.

Q6 - What unusual claims or segments should management investigate?

Identify extreme claim amounts or unusual segment combinations. Do not automatically delete them as outliers; describe them and explain why they may require business review.

Q7 - What should management monitor every month?

Choose the most decision-useful KPIs and design an executive dashboard. Explain why each selected KPI/chart belongs on the dashboard and what action it can support.

Q8 - What are your three data-driven recommendations?

Provide three recommendations using the format Evidence -> Action -> KPI. Recommendations must follow from your analysis and should not imply a causal effect that the data cannot establish.

5. Core KPI Definitions

Policy Count = number of unique policies.

Total Exposure = sum of policy exposure in years.

Claim Count = total number of claims.

Claim Frequency = Claim Count / Exposure. This adjusts claim occurrence for differing exposure periods.

Total Claim Cost = sum of ClaimAmount.

Average Claim Severity = average claim cost among observed claims (state clearly how you aggregate repeated claims per policy).

Do not invent Loss Ratio. Loss Ratio requires premium information, which is not available in the selected freMTPL2 frequency/severity dataset.

6. Python Workflow

Load frequency and severity data → inspect structure and types → quality checks → aggregate claim amounts appropriately → merge by policy ID → create age/vehicle/bonus-malus segments → calculate portfolio and segment KPIs → perform Pareto and outlier review → export clean analysis tables for Tableau.

The Python notebook should show how the final Tableau-ready tables were created. Tableau should not be used to hide cleaning or undocumented manual transformations.

7. Tableau Dashboard Requirements

Create a professional dashboard with 2-3 pages/views rather than many disconnected charts.

Page 1 - Executive Portfolio Overview

KPI cards: Policy Count, Exposure, Claim Count, Claim Frequency, Total Claim Cost, Average Severity. Include at least one segment/region comparison and useful filters.

Page 2 - Claims & Risk Segments

Show driver/vehicle/region patterns. Include a frequency-vs-severity comparison and allow the user to filter or highlight important segments.

Page 3 - Claim Cost Concentration

Show high-cost claims/segments, Pareto concentration and management attention areas. End with three concise business recommendations.

8. Recommended Visuals

Use only visuals that support a decision. Good options include: KPI cards; horizontal bar chart by region/segment; scatter plot of frequency vs severity; distribution of ClaimAmount; Pareto/cumulative cost chart; highlight table by segment; and a region comparison. Avoid decorative charts that do not answer a business question.

Every chart must answer: What do we see? Why does it matter? What decision could this support?

9. Scientific & Business Quality Rules

Normalize claim counts by exposure. A group with more policies or longer exposure naturally has more opportunity to generate claims.

Separate frequency from severity. A segment may claim often but cheaply, or rarely but expensively.

Do not delete large claims automatically. Insurance claim distributions are often highly skewed; investigate first.

Association is not causation. The dashboard supports descriptive decision-making, not causal claims.

Document the merge. Explain how policy-level frequency data and claim-level severity data were linked and aggregated.

Be transparent about limitations. The insurer is anonymous, the dataset is historical, and the available fields do not capture every factor relevant to insurance risk.

10. Required Deliverables

- 1) README.md
- 2) Python analysis notebook (.ipynb) and/or clean .py files
- 3) Tableau workbook (.twb/.twbx) or Tableau Public link
- 4) cleaned Tableau-ready CSV file(s)
- 5) figures/screenshots of the final dashboard
- 6) requirements.txt
- 7) original dataset link and data-source citation
- 8) short final business summary with three recommendations

11. Recommended Repository Structure

```
insurance-claims-dashboard/  
  README.md  
  requirements.txt  
  notebooks/insurance_analysis.ipynb  
  data/processed/tableau_portfolio.csv  
  data/README.md  
  tableau/insurance_claims_dashboard.twb  
  figures/  
  report/business_summary.md
```

12. README Requirements

Business problem; official dataset source; data dictionary; cleaning and merge logic; KPI definitions; Python analysis steps; Tableau dashboard structure; key findings; three Evidence-Action-KPI recommendations; limitations; repository structure; and instructions for reproducing the analysis.

13. Submission Rules - Mandatory

Upload your project on Monday, August 31 or Tuesday, September 1, 2026.

Final deadline: Tuesday, September 1, 2026.

September 2-4 is reserved for feedback, corrections and project-video submission.

ZIP/RAR archives are not accepted. Submit one clean, directly reviewable GitHub project folder/repository.

README.md, Python code and Tableau output are mandatory.

14. What Success Looks Like

A strong project does not merely build an attractive dashboard. It shows that you can prepare insurance data in Python, define correct exposure-aware KPIs, distinguish claim frequency from severity, identify meaningful portfolio patterns, design a clear Tableau dashboard and translate evidence into management decisions.

Suggested GitHub title: Insurance Claims & Portfolio Risk Analytics - Python + Tableau